Indian Institute of Technology Gandhinagar

MA 104: ODE

Semester-II, Academic Year: 2025-26

Tutorial – 5

Assignment. Please submit solutions to Problems 3 and 6. Each problem carries 5 marks. The assignment is due by 11:59 PM on February 11, 2026.

(1) Show that $y_1(x) = x^2$ and $y_2(x) = x^2 \log x$ form a basis of solutions of the homogeneous equation corresponding to

$$x^2y'' - 3xy' + 4y = x^2 \log x, \qquad x > 0.$$

Hence, find the general solution of the non-homogeneous ODE.

(2) Determine the radius of convergence.

(a) $\sum_{m=0}^{\infty} m(m+1)x^m$

(b) $\sum_{m=0}^{\infty} \frac{(-1)^m}{k^m} x^{2m}$

(c) $\sum_{m=0}^{\infty} \frac{x^{2m}}{(2m+1)!}$

(3) Consider the equation $(1 + x^2)y'' - 4xy' + 6y = 0$.

(a) Find its general solution in the form $y = a_0y_2(x) + a_1y_1(x)$, where $y_1(x)$ and $y_2(x)$ are power series solutions in powers of x.

(b) Find the radius of convergence for $y_1(x)$ and $y_2(x)$.

(4) Solve the following equation by means of a power series about the given point $x_0 = 0$. Find the recurrence relation for each of the two linearly independent solutions.

$$(1 - x)y'' + xy' - y = 0, \qquad \text{with} \quad y(0) = -3, \qquad y'(0) = 2.$$

(5) Find a power series solution in powers of x for the differential equation:

(a) $y'' - 4xy' + (4x^2 - 2)y = 0$.

(b) $(1 + x^2)y'' - 4xy' + 6y = 0$.

(c) $(1 - x^2)y'' - 2xy' + 2y = 0$.

(6) Find the first three **nonzero** terms in a power series expansion about $x_0 = \pi$ of the solution to the initial value problem

$$y'' - (\sin x)\, y = 0, \qquad \text{with } y(\pi) = 1,\ y'(\pi) = 0.$$